Keyvan Habibi

Water Engineer | Physics-Informed & Explainable AI for Water Quality | Digital Twins

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RESEARCH PROFILE

Water engineer working on **physics-guided and causal deep learning**, **explainable AI** and **digital twins** for **reservoir water quality** in data-scarce arid regions, with field data from the **Wadi Dayqah Dam reservoir (Oman)**. Co-author with leading researchers in water resources and AI (R. Kerachian, J. Šimůnek, A.H. Gandomi, A. Mishra, R. Nazari). Two years of hands-on operation of a city water distribution system. Available to start a full-time PhD in 2027.

At a glance: 3 Q1 journal articles in 2026 (2 first-author) | 3 first-author manuscripts in preparation | **7 conference papers** | Research Assistant (remote) at **Sultan Qaboos University** since January 2025 | **2 years in a water utility**

EDUCATION

MSc Civil Engineering (Water and Hydraulic Structures), Shiraz University, Iran *2019–2022*

CGPA 15.49/20 (mean of the same-year cohort in the specialisation: 13.73/20); thesis graded **Excellent**; degree awarded 10 September 2022.

Thesis: *Screening environmental impact assessment with a GA/PSO-optimised Takagi–Sugeno–Kang neuro-fuzzy network and metaheuristic feature selection* (NSGA-II, ACO, SA, PSO, GA; Cohen's kappa up to 0.95). Supervisor: Prof. S.H. Afzali.

BSc Civil Engineering, Persian Gulf University, Bushehr, Iran *2014–2019*

RESEARCH EXPERIENCE

Research Assistant, Sultan Qaboos University, Oman *January 2025–present*

Group of Dr Mohammad Reza Nikoo, Department of Civil and Architectural Engineering (remote collaboration)

- **Physics-informed deep learning:** developed a soft-causal ensemble deep neural network for probabilistic water quality assessment in arid regions, trained on 16,096 CTD field measurements from 20 stations (first author, *Journal of Hydrology*, 2026).
- **Digital twin and explainable AI:** built a water quality index with explainable AI and a time-varying 3D digital twin of the Wadi Dayqah Dam reservoir from 31,322 CTD measurement points (first author, *Environmental Development*, 2026).
- **Transferable deep learning:** built a soft-causal, physics-guided deep learning pipeline for chlorophyll-a prediction on 1,012 Mexican inland-water stations and re-ran it on an Omani reservoir, where its predictions broke an assumed temperature-response envelope in 5.5% of nowcast bins, against 20–38% for tree ensembles (first-author manuscript, to be submitted).
- **Water quality indices:** showed that a CCME water quality index recomputed from three of seven indicators matched only 57% of the full index's classes, and designed an evidence-anchored reduced index that matched 97% (with light still profiled) in a stratified supply reservoir (first-author manuscript, to be submitted).
- **Remote sensing:** combined five drone multispectral and thermal surveys, Sentinel-2/Landsat imagery and 501 CTD casts to estimate sub-surface chlorophyll-a with conformal uncertainty; in this reservoir, about 90% of the chlorophyll in the sunlit layer lay below what sensors see (first-author manuscript in preparation).

PUBLICATIONS

Peer-reviewed journal articles

1. **Habibi K**, Nikoo MR, Kerachian R, Nazari R, Šimůnek J, Gandomi AH (2026). A physics-informed, soft-causal ensemble deep neural network for robust probabilistic water quality assessment in arid regions. *Journal of Hydrology* 680, 136466. doi.org/10.1016/j.jhydrol.2026.136466 **(Q1, IF = 7.3)**
2. **Habibi K**, Nikoo MR, Afzali SH, Nazari R, Mishra A, Kerachian R, Šimůnek J, Gandomi AH (2026). Developing a water quality index for reservoirs using explainable AI and a time-varying 3D digital twin: a case study, Wadi Dayqah Dam, Oman. *Environmental Development* 59, 101464. doi.org/10.1016/j.envdev.2026.101464 **(Q1, IF = 5.3, citations: 2)**
3. Karimi A, Nikoo MR, Hoseini Karani MM, Al-Saadi SN, **Habibi K** (2026). Hydro-economic optimization of a novel multi-dam pumped-storage system for energy arbitrage and water-supply reliability in arid regions. *Energy Conversion and Management* 356, 121283. doi.org/10.1016/j.enconman.2026.121283 **(Q1, IF = 11.8, citations: 2)**

Manuscripts in preparation

4. **Habibi K**, Nikoo MR, Shahvaran AR, Kerachian R, Nazari R, Šimůnek J. A transferable soft-causal, physics-guided deep learning pipeline for chlorophyll-a prediction: nutrient effects, calibration and physical consistency in Mexican inland waters and an Omani reservoir. To be submitted to *Water Research*.
5. **Habibi K**, Nikoo MR, Afzali SH, Nazari R, Chen M, Shahvaran AR. Dropping indicators biases the CCME water quality index: evidence-anchored reduction for monitoring a stratified supply reservoir. To be submitted to *Journal of Cleaner Production*.
6. **Habibi K**, Nikoo MR, Shahvaran AR, Kerachian R. How deep can remote sensing see? Drone, satellite, and AI-agent evidence on hidden chlorophyll. In preparation for *Remote Sensing of Environment*.
7. Shahvaran AR, Saravani MJ, Saadatpour M, **Habibi K**. Half a century of inland water trophic state assessment: a review. In preparation for *Journal of Hydrology*.

Conference papers (7 in total; international first-author papers listed)

8. **Habibi K**, Afzali SH. ANFIS against LSTM: precipitation time series forecasting. *6th Int. Conf. on Researches in Science & Engineering & 3rd Int. Congress on Civil, Architecture and Urbanism in Asia, Thailand* (2022).
9. **Habibi K**, Afzali SH. Environmental impact assessment using TSK-FRBS and lookup table. *6th Int. Conf. on Researches in Science & Engineering & 3rd Int. Congress on Civil, Architecture and Urbanism in Asia, Thailand* (2022).
10. **Habibi K**, Afzali SH. Optimizing fire hydrant layout based on uncertainty analysis. *6th Int. Reliability and Safety Engineering Conf., Iran* (2020).
11. **Habibi K**, Afzali SH. Immunization of water distribution networks against terrorist attacks. *6th Int. Reliability and Safety Engineering Conf., Iran* (2020).

PROFESSIONAL EXPERIENCE

Founder and CEO, Tavan Vira Niroo Sazeh, Iran *2026–present*

- Founded and run a contracting company for water supply, canal and utility construction projects.
- All my company activities will be suspended for the full duration of the PhD.

Operation Supervisor, Fars Province Water and Wastewater Company (ABFA), Iran *2023–2025*

- Operated and managed the water distribution system of a city of 11,000 people.
- Supervised field crews in daily inspection of water and wastewater facilities and handled failures.
- Worked on a machine-vision project for automatic water-meter reading.

Construction Intern, Rad Shahr Aftab Inc. and Mohandesin Tarh Saze Inc., Iran *2017–2018*

TEACHING

Teaching Assistant, Shiraz University *2020–2021*

Fluid Mechanics, Hydraulics and Advanced Hydraulics (Prof. S.H. Afzali)

TECHNICAL SKILLS

Machine learning: physics-informed and causal deep learning, ensemble and probabilistic prediction, transfer learning, LSTM, explainable AI, uncertainty quantification

Optimisation and decision: metaheuristics (GA, PSO, NSGA-II, ACO, SA), multi-criteria decision making (fuzzy TOPSIS), neuro-fuzzy systems (ANFIS, Takagi–Sugeno–Kang)

Programming: Python (TensorFlow/Keras, scikit-learn, XGBoost, LightGBM, CatBoost, SHAP, LIME), MATLAB, R

Data: reservoir CTD profiles, GEMStat water quality records, UAV multispectral and thermal imagery, Sentinel-2 and Landsat

HONOURS, AWARDS AND MEMBERSHIP

- Ranked within the top 6% in the Iranian national MSc university entrance examination (2019).
- Ranked within the top 5% in the Iranian national BSc university entrance examination (2014).
- Governmental scholarship for the MSc programme (Shiraz University).
- Certificates of Appreciation from the CEO of Fars Province Water and Wastewater Company (2023, 2024).
- Letter of Appreciation from the Ministry of Energy, Iran.
- Member of the Iranian Construction Engineering Organization.

LANGUAGES

- **English:** fluent; working language of all research, publications and international collaboration. My TOEFL iBT test, booked for 5 September 2026, was cancelled by the test provider in Iran; a score will be provided as soon as a new test can be taken.
- **Persian:** native

REFEREES

Available on request.